Sahaj Bhadja

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Summary

Software Engineer with 2+ years building user-facing **full-stack** applications, **Spark** data pipelines, and production **machine learning** and LLM systems in Python, Java, and JavaScript, using relational and NoSQL databases, Kubernetes, and test coverage.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **data pipelines** with incremental loads, validation, and reconciliation across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs at 99%+ reliability through CI/CD-backed production releases for 500+ users.
- Restructured async LLM orchestration with background workers and failure isolation to triple throughput across 1K+ AI jobs, and tuned indexed queries to cut **dashboard latency** from 1.2s to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **user-facing** React and Flask application on AWS for upload, rubric, and grading workflows, serving 150+ instructors and 10K+ daily requests with stable responses during peak load.
- Integrated multimodal **ML models** including GPT-4o, Claude, and Qwen-VL with OCR fallbacks and per-call usage instrumentation, reaching 90%+ grading consistency and cutting instructor feedback turnaround 60% across 10K+ submissions.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer in **PyTorch** for offline reinforcement learning, with end-to-end data pipelines over 100K+ trajectory logs, improving long-horizon trajectory planning 18% in closed-loop simulation runs.
- Designed a multi-head attention encoder fusing map geometry with agent trajectories, **benchmarking** it against three baselines with Weights & Biases tracking and improving closed-loop planning stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned **ETL** on AWS Glue and S3 that consolidated user activity logs from multiple sources into analysis-ready formats, boosting pipeline efficiency 68% and cutting storage costs 15%.
- Built object-oriented **Java** and Python microservice modules with 20+ Mockito and PyTest suites, and optimized APIs with pagination and caching, improving response times 30% in Agile sprints.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built hourly **Spark** micro-batch pipelines on Databricks over 50K+ booking and payment events, aggregating peak hours, popular services, daily active users, and revenue for internal marketplace dashboards and client reporting.
- Built a React and TypeScript dashboard with Flask APIs and **NoSQL** Firebase real-time sync for a two-sided services marketplace, keeping responses under 200ms and speeding confirmations 40%.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, HTML, CSS
Full Stack: React, Next.js, Node.js, FastAPI, Flask, Spring Boot, REST APIs, Microservices
Big Data: Spark, Databricks, AWS Glue, ETL Pipelines, Data Validation, PostgreSQL, MySQL, MongoDB, Firebase
Machine Learning: PyTorch, TensorFlow, Transformers, LLM Systems, Embeddings, Reinforcement Learning
Infrastructure: Kubernetes, Docker, AWS, CI/CD, Jenkins, Git, PyTest, Jest, Linux

Projects

Job and Candidate Matching Service | *Python, PyTorch, Embeddings, FAISS, FastAPI, Kubernetes*

- Designed a two-stage **matching** service that retrieves candidate jobs with embedding search, re-ranks them with a learned model using skills, title, and location fit, and logs clicks for **offline evaluation**.

Job Alert Event Pipeline | *Spark, Kafka, Python, PostgreSQL, React*

- Designed a **Spark** streaming pipeline that joins new job postings with saved searches, deduplicates alerts per user, and feeds a React dashboard tracking delivery, open, and **apply rates**.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*

- Built a **semantic search** platform with transformer embeddings and FAISS over 50K+ documents, using incremental indexing to cut retrieval latency 40% while sustaining 95% query accuracy for 200+ users.